

Site Name:

Date:

Case/Work Order (WO) #:

Serial #:

Note:

- Items inside the Blue Boxes are to be completed on every 3-Year PM. All other items must be completed at the next scheduled 1-Year PM.
- Unless specified, ALL procedures can be found in the current version of the Panther System Service Manual.
- If some parts are unavailable, document in the Service Report and continue with the PM.
- PMs are not required for internal systems (e.g. Training, R&D, etc.)

This form contains the following sections:

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Complete the following procedures in the order listed. See the current version of the Panther System Service Manual for detailed instructions. Check the box to indicate the step was performed.

Section A: Panther System/Panther ART

Item	Part 1: PC and UPS	Completed (✓)
1	Check with the customer if they have any issues not related to the PM.	☐
2	Review the Messages, Panther Health Report and System Health Report (for at least one week prior) for any unusual activity. <i>If needed, search "Health Reports" in the Panther System Service Manual for detailed instructions.</i>	☐
3	Confirm all assay versions are consistent with the System Software installed. Refer to the respective Technical Bulletin for pertinent information. Delete old version(s) and install new version(s) as necessary.	☐
4	IF Panther ART, perform "Empty Waste" from the Task Icon (unlocking the waste drawer prior to shutting down Panther Software will transfer any liquid waste from the Permanent bottle to the Removable bottle and reset the waste count to 0 thus eliminating Implausible Status of the Permanent Bottle upon Panther Software start up.	☐
5	Ensure Auto Purge of Results is set for Results older than 365 days. If Auto Purge was not previously set up, then set up Auto Purge AND Manually Purge results on the Panther PC. Check with customer to determine date range and only purge after the database has been backed up first. <i>If needed, refer to FS-346 and search "Purging Results Manually on the Panther System PC" in the Panther System Service Manual for detailed instructions.</i>	☐
6	Print a report to verify printer operation.	☐
7	Verify Panther Serial Number is correctly entered in the GUI and in the Computer Name in Windows PC Settings.	☐
8	Delete temporary LIS files (only for systems running System Software v5.3). <i>If needed, refer to FS-523.</i>	☐
9	If Panther ART - from the Panther Shield, login to the Service Shield and select Service. Using the WOTF Transfer pump transfer liquid waste from the Permanent (Static) bottle to the Removable bottle and clear the removable bottle fitting. 1. Click on <i>IC-Waste on the Fly</i> tab in Service. 2. Select the <i>IC-Waste on the Fly</i> in the drop-down list in the Common Control window. 3. Click <i>Initialize</i>. 4. Select the <i>Liquid Waste on the Fly</i> tab. Initialize the Transfer Pump. 5. Add approximately 500mL – 1L of hot water into the Permanent Waste Bottle. 6. Empty the Permanent Waste bottle by turning on the Transfer Pump until there is no fluid left in the clear tubing coming from the bottle. If the fluid in the clear tubing appears cloudy, repeat process until the fluid is clear. 7. With Vacuum turned on – open the Valve. This will allow any liquid in the Removable bottle fitting to be drawn back into the Permanent bottle. (Note: If needed, select the Vacuum and Drawer tab and Activate the	☐

Item	Part 1: PC and UPS	Completed (√)
	vacuum pump.) 8. Empty and discard both the liquid and solid waste. 9. Close out Service.	
10	Backup the Panther Database, including the Dashboard Database.	☐
11	For Panther PC with Windows 7 Image: Perform a defragmentation of the Panther PC or Optimization (Image 6.0.13 and above). <i>If needed, refer to "Defragmentation Procedure on Windows 7" in Panther System Manual.</i>	☐
12	Unplug the UPS from the wall outlet and confirm the UPS switches to battery power. If check fails, troubleshoot and replace batteries if needed (SAP # 717881; Hologic P/N 902722 <u>or</u> SAP # 741904; Hologic P/N PWR-00315, depending on UPS model).	☐
13	While disconnected, thoroughly clean the interior and exterior of the UPS (remove all dirt and debris). Remove the front cover and clean the fan/metallic grill with canned air.	☐
14	Before reconnecting the previously removed plugs to the UPS, make sure that no devices other than the following are connected: • Panther Instrument • Dell PC • Printer • Cisco Firewall If you find any devices besides those four connected to the UPS, please remove them and find another outlet for them, making sure that it is authorized for its use.	☐
15	ALSO complete the following on a 3-Year PM: Replace the UPS batteries. Reset the Input Voltage Ranges. <i>If needed, refer to FS-258 or FS-496.</i>	☐

Item	Part 2: Upper Bay	Completed (√)
Pipettor		
1	Inspect the X-Axis belts for wear, damage, or warping. Replace, if required. <i>If needed, search "Pipettor X-Axis Rack Belts Removal and Replacement" in the Panther System Service Manual for detailed instructions.</i>	☐
2	Check the Sample Bay Cover and Reagent Bay Shutters for leaks or residue from the pipettors. Address any pipettor issues.	☐
3	Remove and clean both sides of the two Reagent Bay Shutters.	☐

Item	Part 2: Upper Bay	Completed (✓)
4	Clean, inspect cLLD gold leads and DiTi. <i>If needed, search "Pipettor DiTi Cleaning" in the Panther System Service Manual for detailed instructions.</i> Note: If the cLLD Washer and DiTi Sleeve upgrade has been performed and there are no customer reported LLS errors, this step is not required.	☐
5	Using a dry, lint-free towel and a small amount of 70% isopropyl alcohol, clean and wipe each DiTi. If it is not already installed, install the new injector sleeve per the Panther System Service Manual.	☐
6	Tighten the DiTi thumbscrews on both the Sample and Reagent Pipettor arms and adjust/tighten, as needed.	☐
7	Inspect Flex Cables for wear and inspect Flex Cable routing.	☐
8	Check Gantry Rails for parallelism. <i>If needed, search "How to Use the Pipettor Rail Alignment Tool" in the Panther System Service Manual for detailed instructions.</i>	☐
9	Using a dry, lint free towel, wipe down all rods, rails and slides of the Pipettor rack and lubricate the gantry rails. <i>If needed, search "Pipettor Gantry Rail Maintenance Procedure" in the Panther System Service Manual for detailed instructions.</i>	☐
10	ALSO complete the following on a 3-Year PM: Replace the Y- and L-Bearings on all pipettors (SAP # 716213; Hologic P/N 902778 and SAP # 716214; Hologic P/N 902780).	☐
Pump Module, MagWash, and Vacuum System		
11	ALSO complete the following on a 3-Year PM: Clean the Vacuum Manifold Insert and the inside of the Vacuum Manifold. Replace the manifold insert O-ring (SAP # 740704; Hologic P/N MME-02597) and if needed, replace the manifold O-rings (SAP # 740703; Hologic P/N MME-02931). Apply High Vacuum Grease (SAP # 716265; Hologic P/N 903524) to manifold insert. <i>If needed, search "How to Install the Vacuum Manifold Insert" in the Panther System Service Manual for detailed instructions.</i>	☐
12	Clean the Vacuum Manifold Pressure Sensor Port. <i>If needed, search "Vacuum Manifold Pressure Sensor Port Cleaning" in the Panther System Service Manual for detailed instructions.</i>	☐
13	Check all the fittings on the Vacuum Manifold. Replace any broken Vacuum Manifold fittings.	☐
14	Check ALL MagWash 1 and MagWash 2 valve manifold fittings.	☐
Pump Module, MagWash, and Vacuum System (Continued)		
15	Inspect the MagWash drip tray and all fluid connections on the MagWash (Oil, Wash and Aspirators).	☐

Item	Part 2: Upper Bay	Completed (✓)
16	Check ALL MagWash Aspirators are on firmly (use a 7 mm wrench – DO NOT overtighten).	☐
17	Check there are no crimps in the MagWash aspirator tubing.	☐
18	Check the Pump Module drip tray for leaks.	☐
19	Check for leaks, cracks, and kinked tubing on the Wash, Auto Detect, and Oil pumps. DO NOT overtighten inlet and outlet fittings.	☐
20	Replace the Peristaltic Pump Tubing as needed Use (SAP # 716767; Hologic P/N 902777) + (SAP # 740051; Hologic P/N MME-02391) + [(SAP #743055; Hologic P/N MME-04964) + (SAP #740598; Hologic P/N MME-02761)]. Write the date of replacement on a piece of tape and affix to the pump.	☐

Item	Part 3: Mid Bay	Completed (✓)
Output Queue Part 1		
1	Clean all bleach residue from the entire Output Queue.	☐
2	Remove the Output Queue lid and clean with bleach (0.5-0.7% sodium hypochlorite) and then DI water.	☐
3	Clean and LIGHTLY oil the Hook Drive and Actuator Drive rails.	☐
4	Adjust the Hook Drive Belt Tension 1. Loosen the 2 screws with the green lacquer to let the spring push out the pulley. 2. Tighten the top screw first and then the bottom screw.	☐
5	Check the integrity of the Red Aspirator tubing [903299]. If brittle or worn replace.	☐
6	Clean deposits on the outside of the aspirators: 1. Manually pipette 4 mL of warm water or Endozime into each tube of three MTUs. 2. With the instrument turned off, manually place the MTUs into the Output Queue Stack. 3. Push MTUs to end of stack and manually lower the aspirators into the first MTU. Each aspirator must be immersed. 4. Continue with the rest of the PM while the aspirators are soaking. Output Queue maintenance is completed in Part 5.	☐
7	Reinstall the Output Queue lid. Verify the lid is seated properly on the stanchion and the O-ring is present. If the O-ring is missing it is only available with the OQ Lid (SAP # 740501; Hologic P/N MME-02534).	☐

Item	Part 3: Mid Bay	Completed (✓)
8	Inspect and clean the Output Queue injectors. If there are any leaks, residue buildup, cracks, or any visible damage, replace the injector (SAP # 716269; Hologic P/N 903641).	☐
9	Apply grease to the injector housing before re-installing the injector. Do NOT block path of fluid stream. <i>If needed, search "Cleaning and Applying Grease to the Output Queue Injector" in the Panther System Service Manual for detailed instructions.</i>	☐
10	Adjust the Hook Drive Belt tension.	☐
11	Reinstall the Output Queue lid. Verify the lid is seated properly on the stanchion and the O-ring is present.	☐
12	ALSO complete the following on a 3-Year PM: Replace the OQ Injector (SAP # 716269; Hologic P/N 903641) and the OQ Injector Housing (SAP # 717229; Hologic P/N 903647). Replace Aspirator Assembly (SAP # 716235; Hologic P/N 902933) and Barb Fittings (SAP # 740384; Hologic P/N MME-02698) that connect to the Vacuum Manifold.	☐ ☐
Sample and Reagent Tip Eject Chutes – Panther Classic		
13	Inspect and clean the Sample and Reagent Tip Eject Chutes. <i>If needed, search "Inspecting and Cleaning the Tip Eject Chute" in the Panther System Service Manual for detailed instructions.</i>	☐
14	Inspect and clean the Sample Dispense Slot.	☐
Tip Carousel and Triplet Catcher – Panther ART		
15	Inspect and clean any Tips, Triplets, or MTUs from the Triplet Catcher Splash Shield.	☐
16	Inspect and clean Tip Carousel and Waste Chute.	☐
Load Stations		
17	Check all load stations for loose screws. Tighten if necessary.	☐
18	Check all load station doors open and close smoothly. <i>If needed, clean guide rods with lint-free cloth and 70% ethanol.</i>	☐
MagWashers		

Item	Part 3: Mid Bay	Completed (√)
19	Inspect MagWash Aspirator Drive lead screws for excessive or no grease. This is visible on MagWash 2 without the need to remove the assembly. For MagWash 1 the aspirators will need to be in the Aspirate position – inspection of the drive can be done through the top of the assembly. 1. If excess grease is suspected or found, the MagWashers will need to be removed for cleaning. 2. If no grease is apparent on initial inspection – it may be applied without removing the modules to the screw below the motor coupling and above the bearing (aspirators in Aspirate position). Grease (Super Lube or comparable) can be applied with an applicator stick, (gloved) fingertip, or anything that can deliver a small amount of grease a. Using Service Software exercise the aspirator Drive up and down several times to distribute the grease (can be done at the end of the PM during the Service Software Verification).	☐
Incubators		
20	Check all Incubator doors for door clips or screws and install door clips (SAP # 716072; Hologic P/N 903551) if needed. <i>If needed, search "Door Clip" in the Panther System Service Manual for detailed instructions.</i>	☐
21	Check home pin is not scraping the heater coil frame.	☐
22	Check that all Incubator doors open and close smoothly. <i>If needed, clean guide rods with lint-free cloth, 70% ethanol and lubricate with Standard Lubrication Kit (SAP # 717920; Hologic P/N 903864).</i>	☐
23	Inspect all Incubator pulleys, bearings, and insulation for wear or damage.	☐
24	Inspect all Incubator Belts (SAP # 740942; Hologic P/N MME-03197) for wear and replace as necessary.	☐
Incubators (Continued)		
25	Inspect External Amp Incubator fan for dust. Remove and clean if necessary. Reinstall the fan after cleaning.	☐
26	Replace the High Temperature Incubator Fan (SAP # 716268; Hologic P/N 903627).	☐
Input Queue – Panther Classic		
27	Check the IQ Ball Detent and adjust if necessary.	☐
28	Lubricate the IQ Ball Detent and check the IQ Drawer for smooth movement. Use Linear Bearings Grease (SAP # 741459; Hologic P/N MME-02134). -Inspect the IQ Ball Detent Sled and tighten if needed. -Inspect the Bumpers to make sure they are screwed all the way. -Perform the IQ Belt Tensioning -Tighten the screw	☐

Item	Part 3: Mid Bay	Completed (√)
29	Check the Input Queue for loose mounting screws and check lock washers on MTU backstop. <i>If needed, search "Input Queue Screw Fastening and Inspection Procedure" in the Panther System Service Manual for detailed instructions.</i>	☐
30	If the Health Report or Logs shows LD step loss errors at the IQ, check the Input Queue rail alignment. <i>If needed, search "Input Queue MTU Rail Alignment Procedure" in the Panther System Service Manual for detailed instructions.</i>	☐
MTU Queue – Panther ART with MTU Expansion Module		
31	Clean the MTU Queue.	☐
32	Perform the MTU Queue Belt Tensioning 1. Move the Linear Distributor away from the MTU Queue 2. From the front left of the Queue loosen the screw on the belt pulley one turn and the belt will self-tension 3. Tighten the screw	☐
MTU Sidecar – Panther ART with MTU Expansion Module		
33	Inspect all carriers for bent bases and bent or misaligned rails.	☐
34	Clean and remove any debris from the MTU Drawer.	☐
35	Inspect MTU Sidecar MTU Drawer door for misalignment.	☐
Linear Distributor		
36	Check the area around all modules for MTU shavings indicating an alignment issue with hook movement or teaching.	☐
37	Thoroughly clean the entire distributor module (including the rails) of any dirt or debris.	☐
38	Check the entire module for loose screws, especially the distributor head, and tighten if necessary. <i>Refer to FS-299 and search "Updated Linear Distributor Model Information" in the Panther System Service Manual for a complete list of hardware to inspect.</i>	☐
39	Check for signs of damage to the distributor head or on other modules indicating the head is contacting another module during operation.	☐
40	Inspect all distributor ribbon cables (x-axis, theta, and hook teach sense cables) for damage. Replace, if damaged.	☐
41	Inspect all motor and encoder cables for damage/abrasions or loose connectors at the boards and motors. Replace any component, as necessary.	☐
Linear Distributor (Continued)		
42	Verify theta drive gear meshing is not too tight or too loose. Use a piece of paper as a guide for meshing.	☐

Item	Part 3: Mid Bay	Completed (√)
43	Clean all teach pins, the hook, and the hook-grounding tab to ensure good contact during auto-teach.	☐
44	Ensure the hook-grounding tab is bent toward the hook to properly ground for teach-sense calibration. Apply DeOxit to the contact point between the hook and the grounding tab.	☐
Luminometer		
45	Remove the injector, clean with DI water, and reinstall. Note: Ensure the depth of the DI water does not go above the injector site holes.	☐

Item	Part 4: Lower Bay	Completed (√)
Cooling Module		
1	Check Cooling Module is mounted securely to the base plate.	☐
2	Check all foam pieces are seated properly, specifically the piece that holds the PCB.	☐
3	Inspect cooling lines and fittings for leaks and kinks. Repair as needed.	☐
4	Check the Mid Bay Drawer does not close on any lines.	☐
5	ALSO complete the following on a 3-Year PM: Replace the Cooling Module Pump's Diaphragm and Valves (SAP # 741096; Hologic P/N MME-03295). <i>If needed, search "Cooling Module Pump Diaphragm and Valves Removal and Replacement" in the Panther System Service Manual for detailed instructions.</i>	☐
Vacuum Assembly		
6	Check the tubing at the bottom of the Vacuum Housing and trim if any cracks or splits are found.	☐
Universal Fluids Drawer		
7	Label (A or B) and remove all bottles and place them on a bench pad.	☐
8	Replace any bottle caps, straws, and connections, as needed. <i>If needed, search "Bottle Straw Connections" in the Panther System Service Manual for detailed instructions.</i>	☐
9	ALSO complete the following on a 3-Year PM: Replace the Deactivation Buffer Bottle Straws with Fitting (SAP # 740759; Hologic P/N ASY-09872).	☐
10	Replace the Deactivation Buffer Bottle's CPC fittings (SAP # 715976; Hologic P/N 902775) on sites where each Panther is processing about 2000 tests or more per week.	☐
11	Check the float inside the Bleach Bottle for wear, damage, or discoloration. A common sign of damage is the presence of black particles floating in the bottle. Replace the bottle if the float is damaged.	☐

Item	Part 4: Lower Bay	Completed (✓)
12	Clean all bottle fittings using paper towels or Kimwipes and DI water. Use different towels for different bottles.	☐
13	Check ventilation holes in bottle caps for clogs and clear if necessary.	☐
14	Inspect all fluid lines throughout the module for any signs of leaks, spills, or kinks. Clean any spills and repair all leaks.	☐
15	Clean the interior surfaces of the drawer.	☐
16	Check all solenoid valves for leaks, replace if needed. Tighten all fittings.	☐
17	ALSO complete the following on a 3-Year PM: Replace the Wash Buffer Solenoid Valve (Use SAP # 716223; Hologic P/N 902844 to exhaust, then use SAP # 740919; Hologic P/N MME-03043 and SAP # 740920; Hologic P/N 903300). Note: Ensure that the model number of the valve is facing to the left of the system.	☐
18	Replace the Deactivation Buffer Solenoid Valve [Use SAP # 716223; Hologic P/N 902844 to exhaust, then use SAP # 740919; Hologic P/N MME-03043 (Valve only) and SAP # 740920; Hologic P/N 903300]. Note: Ensure that the model number of the valve is facing to the left of the system.	☐
19	Replace the Deactivation Buffer in-line fitting (SAP # 741529; Hologic P/N MME-04039).	☐
20	ALSO complete the following on a 3-Year PM: Replace the Bleach bottle's CPC fittings. (Use SAP # 716254; Hologic P/N 903301)	☐
21	Wipe the Bleach Bottle with bleach (0.5-0.7% sodium hypochlorite) followed with a DI-water rinse.	☐
22	Refill the Bleach Bottle with bleach (5-8.25% sodium hypochlorite) and reload onto the Universal Fluids Drawer.	☐
23	Return all Fluids Bottles to the Universal Fluids Drawer.	☐
24	Ensure the Universal Fluids Drawer's magnetic lock is properly aligned, the drawer closes smoothly and is flush with the Mid Bay.	☐
25	Open the Waste Drawer, empty and discard both the liquid and solid waste accordingly.	☐
26	Clean the Waste Drawer Bay and surrounding areas. Remove all debris.	☐
Waste Drawer – Panther Classic		
27	Inspect the fittings under the Waste Mounting Block for cracks and damage. If it is not installed yet, install the new Waste Bottle Mounting Block (SAP # 741136; Hologic P/N MME-03428). Note: Check the new or existing block for the latest version of the manifold fittings and replace as needed (SAP # 741651; Hologic P/N ASY-15115). <i>Refer to FS-488.</i>	☐

Item	Part 4: Lower Bay	Completed (✓)
28	Replace the Waste Bottle O-rings [SAP # 715973; Hologic P/N 902236 (use to exhaust first) or SAP # 741845; Hologic P/N MME-04282] and apply High Vacuum Grease (SAP # 716265; Hologic P/N 903524) to the O-rings. <i>Refer to FS-564.</i>	☐
29	Check the Waste Bottle for leaks and cracks. Pay close attention to the seals. If leaks or cracks are found, replace the Waste Bottle.	☐
30	Check the Waste Bottle Magnets – if magnets are loose, misaligned or corroded, replace the Waste Bottle.	☐
31	Inspect the elbow fitting under the rear left corner of the Waste Drawer for cracks and damage.	☐
32	Inspect all Vacuum tubing for kinks.	☐
33	Replace the Vacuum filter. T8 Filter (SAP # 715985; Hologic P/N 903401) to exhaust then use Odorcarb Filter Cartridge (SAP # 741538; Hologic P/N ASY-11890). Ensure that the flow arrow points up. <i>Refer to GDSS-QA-TB-000043 if necessary.</i>	☐
34	Ensure the Waste Drawer's magnetic lock is properly aligned; the drawer closes smoothly and is flush with the Mid Bay.	☐
Waste Drawer – Panther ART: On-demand Access Waste Management Module		
35	Open the Waste Drawer and inspect the permanent Liquid Waste Container and ensure there is no residue buildup.	☐
36	Clean the Waste Drawer Bay and surrounding areas. Remove all debris.	☐
37	Inspect both the removable and permanent Liquid Waste Containers for any damage. - Inspect the inside of the removable Liquid Waste Container. If any corrosion is observed replace the bottle (SAP # 741292; Hologic P/N MME-03754). - Inspect the Coupling Block Fitting on the removable Liquid Waste Container for cracks – replace the bottle if necessary (SAP # 741292; Hologic P/N MME-03754). - Replace the removable Liquid Waste Container O-ring (SAP # 741430; MME-04014) and apply High Vacuum Grease (SAP # 716265; Hologic P/N 903524) to the O-ring. - Inspect the fittings of the permanent Liquid Waste Container. - Replace the permanent Liquid Waste Container screw cap and gasket (SAP # 741618; Hologic P/N MME-04096).	☐
38	Inspect all tubing for wear, stiffening or discoloration. Replace tubing where deemed necessary.	☐
39	Inspect the elbow fitting at the rear of the Waste Drawer for cracks and damage.	☐
40	Inspect all Vacuum tubing for kinks.	☐

Item	Part 4: Lower Bay	Completed (✓)
41	Replace the Vacuum filter – Odorcarb Filter Cartridge (SAP # 741538; Hologic P/N ASY-11890). Ensure the flow arrow is facing up. <i>Refer to GDSS-QA-TB-000043</i>	☐
42	Inspect the Vacuum Filter Block and clean as needed.	☐
43	Replace the Bellows Assembly on the WOTF Transfer Pump (SAP # 743233; Hologic P/N MME- 05086).	☐
44	Replace the Solenoid Valve (SAP # 741479; Hologic P/N MME-04053) on the removable Liquid Waste Container Coupling Block.	☐
45	ALSO Complete the following on a 3-Year PM: Replace the male fitting (SAP # 741616; Hologic P/N MME-04094) on the bottom of the removable Liquid Waste Container Coupling Block.	☐
46	Ensure the Waste Drawer's magnetic lock is properly aligned; the drawer closes smoothly and is flush with the Mid Bay.	☐
Waste-to-Drain Module (If Applicable)		
47	Inspect the tube routing from the Waste Drawer to the Waste-to-Drain Module. Make sure it is not kinked.	☐
48	Inspect the removable Liquid Waste Container fitting for Waste-to-Drain. If dry and brittle, replace the fitting (SAP # 741490; Hologic P/N MME-03763) or connection assembly (SAP # 741391; Hologic P/N ASY-13736).	☐
49	Check for buildup of residue and rinse out the removable bottle before installing.	☐
50	Inspect the customer drain for any obstructions or clogging.	☐
Power Supply		
51	Ensure Power Supply is clean and dust free	☐
Miscellaneous		
52	For LCD/LED monitors, check and tighten the monitor gimbal and arm bolt, as needed.	☐
53	Inspect the Canopy Door air springs (SAP # 716246; Hologic P/N 903079) and replace as needed. <i>If needed, search "Inspecting and Replacing Canopy Door Shocks" in the Panther System Service Manual for detailed instructions.</i>	☐
54	ALSO complete the following on a 3-Year PM: Replace the Canopy Door air springs (SAP # 716246; Hologic P/N 903079).	☐
55	Inspect the Canopy Door microswitches (SAP # 716231; Hologic P/N 902921) and ensure they are not broken or loose. Replace them, if necessary.	☐
56	Clean the glass on all scanners (Reagent Bay, Sample Bay, TCR, and MTU). Use 70% ethanol, followed by DI water and a lint-free cloth. Make sure the glass is dry.	☐

Item	Part 4: Lower Bay	Completed (✓)
57	Clean the Sample Bay Cover for warping. If the cover is warped, check if customer has non-warped spare. Swap cover and order replacement cover (SAP #741946; Hologic P/NMME-04344). Discard warped cover. <i>If needed, search "Sample Bay Cover and Pipettor Interference Check" in the Panther System Service Manual for detailed instructions.</i> Note: Sample Bay Lid, Upgrade Kit (SAP #741945; Hologic MME-04343) includes 741946)	☐
58	Check the screws on the 24V Power Supply and tighten, if necessary. <i>If needed, search "Check 24V Power Supply Screw Connection" in the Panther System Service Manual for detailed instructions.</i>	☐
59	Inspect both sets of chassis fans on either side of the instrument for dust. Remove and clean if necessary. Reinstall fans after cleaning.	☐

Item	Part 5: Panther Service Software Verification	Completed (✓)
Turn on the instrument, log in to the FSE Shield, and start Panther Service Software.		
Linear Distributor		
1	IF the Distributor Head position or Theta Gear Meshing was readjusted, reteach the distributors at all modules. <i>If needed, search "Distributor Alignment and Teaching Procedure" in the Panther System Service Manual, then refer to "Distributor Manual Teaching to Any Module" for detailed instructions.</i>	☐
MagWashers		
2	If any cleaning and/or lubricating of the Aspirator Drives was performed without removing the modules exercise the motor up and down several times to distribute the grease on the lead screw. If either or both Magwashers were removed perform a MagWash OQ on either or both.	☐
Pipettor		

Item	Part 5: Panther Service Software Verification	Completed (✓)
3	IF gantry parallelism was readjusted, or IF V/L bearings replaced on the 1-year PM, IF Pipettors or Pipettor belts were replaced, or IF any module the pipettors interact with were removed and reinstalled, or IF the Pipettor Gantry was raised, reteach Pipettor alignment. <i>If needed, search "Pipettor Teaching" in the Panther System Service Manual for detailed instructions.</i>	☐
Output Queue Part 2		
4	Initialize the Output Queue.	☐
5	When prompted, initialize and activate the Vacuum Pump (click YES).	☐
6	Observe the aspiration of the fluid from the MTU placed in Section A, Part 3, Output Queue Part 1 . - During aspiration, carefully observe the red aspirator tubing and confirm that all fluid is aspirated uniformly. - If the speed of aspiration is NOT uniform across all tubes, manually pipette 4 mL of Endozyme into each tube of an MTU and aspirate again. - If the aspiration is still not uniform, replace Aspirator Assembly (SAP # 716235; Hologic P/N 902933) and Barb Fittings (SAP # 740384; Hologic P/N MME-02698) that connect to the Vacuum Manifold.	☐
Vacuum and Cooling System		
7	Initialize and activate the Liquid, Vacuum and Cooling System.	☐
Vacuum and Cooling System (Continued)		
8	Drain and refill the Cooling Module. Verify the cooling tubing on the back of the Sample Bay is routed properly.	☐
9	Check the Vacuum Pumps are operating within -203 mBar (6 inHg) to -270 mBar (8 inHg).	☐
10	Confirm the Sample Bay, Reagent Bay, and Cooling Module temperatures are within range.	☐
11	Confirm the Sample Bay, Reagent Bay, and Cooling Module temperatures are within ~1.5°C of each other when at a steady state.	☐
12	If the Vacuum is noisy, check the Vacuum muffler attachment.	☐
Incubators		
13	Confirm smooth MTU travel in and out of the incubator.	☐
14	Check for Step Losses. Realign IF necessary. If the Health Report or Logs shows LD step loss errors at Incubator – check and adjust Slot 1 alignment. If any slot is realigned, reteach the Distributor to Incubator.	☐
Panther ART (If Applicable)		
15	Perform a 2 cycle On-demand Access Waste Management Module OQ.	☐

Item	Part 5: Panther Service Software Verification	Completed (✓)
16	If MTU Expansion Module is installed, perform the MTU Expansion OQ. Note: If the MTU Expansion Module was unfused for any reason, reattach the Sidecar Scissor prior to performing the MTU Expansion OQ.	☐
System Level Operational Qualification (OQ)		
17	Complete a 2-cycle System Level OQ with all modules. Watch the MTU pick and place quality at each location and listen for any loud mixers in the Load Stations and MagWash assemblies. Replace the MTU carrier [ASY-10059] if necessary.	☐
18	Verify smooth movement of the MTU in and out of slots. If movement is not smooth, reattach, as necessary. <i>If needed, search "System Level Operational Qualification (OQ)" in the Panther System Service Manual for detailed instructions.</i>	☐

Section B: General Cleaning & Maintenance

Item	General Cleaning & Maintenance	Completed (✓)
Refer to Panther System Service Manual > Cleaning Procedures.		
1	Panther ART: Prior to launching the Panther GUI, empty the bottle via Panther Service Software until there is no fluid left in the clear tubing coming from the permanent Liquid Waste Container.	☐
2	Clean all tips, Tiplests, MTUs, and other debris from inside the instrument.	☐
3	Clean and dust-off covers, canopy, and panels.	☐
4	Clean and dust the upper deck in and around the TCR Carousel.	☐
5	Ensure the area around the PC is clean and dust-free.	☐
6	Ensure all cables are routed away from the Waste Drawer rails and lock.	☐
7	Ensure the area around the UPS is clean and dust-free.	☐
8	Wipe all external surfaces with lint-free cloth.	☐
9	If the Workstation is a Dell XE3 replace the CMOS Battery [CMP-02047] as needed. <i>Refer to GDSS-RD-TB-000002</i>	☐

Section C: System Verification

Item	System Verification	Completed (✓)
GUI Check		
1	Verify Panther Main starts with no errors.	☐

Item	System Verification	Completed (√)
2	Reset ALL Waste Drawer's solid and liquid waste count.	☐
3	From the Tasks screen, select Prime Instrument , and verify prime passes. <i>Note: Pay particular attention to any abnormal noises-e.g. squeaky MagWash Aspirator drive.</i>	☐
Panther System Check		
4	Perform Performance Qualification (PQ) Procedure. <i>Refer to 290989.</i>	☐
5	For systems running Procleix Parvo/HAV Assay or Procleix UltrioPlex E Assay, complete FlashCheck on all PMs (1-Year and 3-Year PM). <i>If needed, search "Flashcheck - Lumo Injector Verification Procedure" in the Panther System Service Manual for detailed instructions.</i>	☐
6	Complete the following ONLY on a 3-Year PM: Perform the SysCheck Procedure. (If you run both SysCheck and FlashCheck, always complete SysCheck first). <i>If needed, search "SysCheck - Lumo PMT Calibration Procedure" in the Panther System Service Manual for detailed instructions.</i>	☐

Section D: Release Instrument to Customer

Item	Return Instrument to Customer	Completed (√)
1	If any fluids were used in Panther Service Software while servicing the instrument, replace the fluid set(s).	☐
2	Check the onsite supply of Syscheck and FlashCheck reagent kits and restock, if necessary. Sites running Procleix Parvo/HAV Assay, Procleix ArboPlex or Procleix UltrioPlex E Assay should have at least 1 unexpired box of FlashCheck available.	☐
3	Ensure all reagents, panels, calibrators, and controls are not expired. Discard any expired material.	☐
4	Inventory the customer's spare parts list and order parts if necessary.	☐
5	Return the keyboard to the PC Bay and ensure cables do not interfere in the closing of the Waste Drawer.	☐
6	Clean all tools used in the PM procedure. <i>If needed, search "Cleaning Tools" in the Panther System Service Manual for detailed instructions.</i>	☐

Comments/Corrective Action (record attachments if necessary):

Reference	GDSS-RD-DHM-000198-FORM-01	Version	3.0	Status	Effective	Effective Date	06-Apr-2026	18 of 18
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Grifols Representative		Customer	
Printed Name		Printed Name	
Signature	Date	Signature	Date